Statement of Work (SOW) for AME Procurement

A requirement of the INRODES project award is to partner with an additive manufacturing (AM) supplier that is able to explicitly meet the production requirements for powder bed fusion components. Advanced Manufacturing and Engineering (AME) has been identified as the only possible partner within the US to achieve this. The National Aeronautics and Space Administration (NASA), specifically Marshall Space Flight Center (MSFC) has a need for fabrication and post-processing of additively manufactured rotating detonation rocket engine hardware. Specifically, combustion chambers and injectors designed for the 10K lbf class. This hardware must be designed and fabricated with no witness lines, restarts, or coater arm tears. Moreover, all passages must be free of powder. All parts must be post processed in a specific order and with specific processing criteria. AME is currently the only potential industry partner with these processing requirements. Thus, a sole source selection of this work is being submitted for procurement.

Vendor:

Advanced Manufacturing and Engineering (AME)
128 Hall Bryant Circle
Huntsville, AL 35806

Task 1, Production of Additively Manufactured Hardware

AME shall produce the hardware listed in the provided Quote. All hardware will be designed by NASA engineers at Marshall Space Flight Center and their respective models supplied to AME before production. During production AME must meet the following requirements. Provide all CORE hardware with no witness lines, restarts, or coater arm tears. Provide all hardware with no powder blockages.

Task 2, Post Processing of Additively Manufactured Components

Several of the required post processing steps shall be conducted by AME. All hardware shall be de-powdered, and no passages shall be blocked with powder. Parts will be sent to a mutually agreed vendor for Hot Isostatic Pressing (HIP) operations. AME will remove the part from the build plate using electrical discharge machining (EDM). Vacuum nucleation will also be conducted to clean the parts.

Requirements for Additively Manufactured GRCop-alloy hardware:

1. Vendor shall fabricate all chamber and injector hardware using powder bed fusion (PBF) additive manufacturing.
2. Vendor shall provide material to complete all builds.
3. Density of the material in the as-built condition shall be a minimum of 99.2% prior to any post-processing heat treatments or Hot Isostatic Pressing (HIP) operations and be demonstrated prior to fabrication of any hardware.
4. Vendor shall complete build in one-piece and remove all powder from the channels prior to further operations defined in Requirement #6.
5. Part shall not be exposed to moisture until powder from channels can be confirmed.
6. Part shall complete Hot Isostatic Pressing (HIP) according to NASA processing parameters. Part shall remain on the build plate for HIP operations to help with handling.
7. As-built surface finish requested, no media blasting.
8. Any anomalies during the build process of the final part should be reported to MSFC Technical Point of Contact.
9. Vendor shall remove parts from the build plate prior to delivery.
10. Parts shall be delivered to Marshall Space Flight Center.

11. Vendor shall provide a basic build package including any build anomalies and data relative as evidenced of process validation under documentation required by AS9100, ISO9001, or similar standard.
12. Vendor shall be ITAR registered or demonstrate ability to handle ITAR data.

Total Cost of Hardware:

Hardware shall be procured per the supplied quote from AME for under this procurement. The WBS

Shipping

1. Contractor shall ship any applicable hardware deliverables to:

Acronyms:

CT = Computer Tomography
EDM = Electrodischarge Machining
GRCop = Glenn Research Center Copper-Alloy (Cu-Cr-Nb)
HIP = Hot Isostatic Pressing
ITAR = International Traffic in Arms Regulations
MSFC = Marshall Space Flight Center
NLT = No Later Than
PBF = Powder Bed Fusion